Chapter 1 Introduction: What is Syzygy?

Notions:

1.1 Hilbert's Theorem on Syzygies

† **Problem 1.1.1** *Let: $S = \mathbb{C}[z_0, \ldots, z_r]$ in $r+1$ variables over the complex numbers. Let E be a finitely generated graded S-module. Choose homogeneous generators:*

$$m_1, \ldots, m_b \in E$$

with $deg(m_i) = a_{0,i}$. These determine a surjective mapping

$$\bigoplus S(-a_{0,j}) \xrightarrow{\epsilon} E \to 0 \qquad (f_1, \ldots, f_b) \mapsto \sum f_i \cdot m_i$$

Then, choose $ker(\epsilon)$ and continue apply the method above, we build a sequence

$$\ldots \longrightarrow P_2 = \bigoplus S(-a_2, j) \longrightarrow P_1 = \bigoplus S(-a_{1,j}) \longrightarrow E \longrightarrow 0$$

If at each stage we choose the generators minimally in a suitable sense, we can arrange that the matrices defining the δ_i not have non-zero constant entries. In this case: It is elementary that $P_\bullet$ is unique up to ismomorphism.

▶ ***Theorem*** (Hilbert) The process just described terminates after at most $r+1$ steps.
In other words, any finitely generated S-module E has a minimal graded free resolution of length at most $r+1$, up to isomorphism.

Remark Syzygies and Varieties:
Notice that, the base ring S is Noetherian. Any variety can be represented as an ideal generated by finite numbers of generators,

$$X = V(f_1, \ldots, f_r)$$

Then, consider the syzygy of

$$\bigoplus S(-d) \to A(X) \to 0$$

♣ **Example 1.1.0.1** (Resolution of the Twisted Quartic Curve) Consider the following curve

$$\mathbb{P}^1 \hookrightarrow \mathbb{P}^3 \qquad [s, t] \mapsto [s^4, s^3 t, s t^3, t^4]$$

Take the ideal $I_C \subseteq S = \mathbb{C}[x, y, z, w]$ of C, generated by a quadric and three cubics:

$$Q = yz - xw, \begin{cases} F_1 = z^3 - yw^2 \\ F_2 = xz^2 - y^2 w \\ F_3 = y^3 - x^2 z \end{cases}$$

Then, the module of Syzygies among these generators is spanned by the four relations:

$$\begin{cases} z^2Q - yF_1 + wF_2 = 0 \\ ywQ - xF_1 + zF_2 = 0 \\ xzQ - yF_2 - wF_3 = 0 \\ y^2Q - xF_2 - zF_3 = 0 \end{cases}$$

† **Problem 1.1.2** *Is there any structure to the toatlity of all resolutions of some large class of S-modules.*

There are very few modules $E$for which one can write down the resolution $P_\bullet$ explicitly. Instead, much of our attention will be fucused on the question relating to the **grading** of $P_\bullet$. This involves two sorts of invariants.

Definition 1.1.0.2 The **(graded) Betti numbers** $b_{i,j} = b_{i,j}(E)$ are dsefined by writting

$$P_i = \oplus_j S(-j)^{b_{i,j}}$$

with $\beta_{i,j}$ numbers in degree of j.

Another definition of Betti numbers is given in algebraic topology. Those definitions coincide in Homological algebra.

† **Problem 1.1.3** *Which Betti numbers $b_{ij}(E)$ are nonzero?*

Definition 1.1.0.3 Define:

$$reg(E) := \max\{j - i | b_{i,j\}(E) \neq 0}$$

Then, $reg(E) \leq m \Leftrightarrow E$ is generated in degrees$\leq m$ and for every $i$the ith module of syzygies of E is generated in degrees$\leq i + m$.

Chapter 2 Hilbert's Syzygies Theorem,Free Resolution and Hilbert Functions

2.1 The Study of Syzygies

2.1.1 Hilbert's Functions

Definition 2.1.1.1 Let:

- Space $\mathbb{P}^r := \mathbb{P}^r_K$,$S = K[X_0, \ldots, X_r]$;
- Let $M = \oplus M_d$ be graded S-module, which is finitely generated;
  Then, M_d is a finite-dimensional vector space over K. (Each S_d is finite-dimensional and M_d is finitely generated)
- Define:

$$H_M(d) := dim_K(M_d)$$

2.2 Examples of Free Resolutions

▶ ***Theorem*** Let $v_1, \ldots, v_{ds}$

- **Proof**

□

2.3 The Local Setting

Fix a Noetherian local ring $(A, \mathfrak{m})$ of dimension n.

2.4 Some Local Invariants

2.5 Graded Syzygies

Chapter 3 Introduction to Boij-Söderberg Theory